Replication Report:
Current Driven Magnetization Dynamics in Helical Spin Density Waves

arXiv:cond-mat/0511224 (Wessely, Skubic, Nordström; PRB **73**, 144431 (2006))

REPLICATE-PROJECT / TEXTURES-100 automated replication

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Classification audit (up front)

This paper was filed under the **loop-current** texture class. It is **not** a loop-current / kagome flux-phase paper: there is no orbital current, no Peierls flux, no kagome lattice, and no time-reversal-breaking kinetic term. The physics is **spin-transfer torque (STT) driven magnetization dynamics in a helical spin density wave (spin spiral)**: a charge current along the spiral axis transfers spin to the local moments and rigidly rotates (”slides”) the spiral. The shared kagome loop-current kernel (`loop_current_kagome_kernel.py`) is therefore **not applicable** and was **not imported**. We replicate the actual in-scope core with a purpose-built spin-spiral tight-binding model. *This is a taxonomy mislabel in the input set, reported honestly, not a physics failure.*

1 Paper summary

The authors propose that a DC charge current flowing along the axis of a helical spin density wave (SDW) exerts a bulk spin-transfer torque that rigidly rotates the spiral. Using semiclassical Boltzmann linear response (their Eqs. 3–6) combined with a full-potential APW+lo LSDA calculation of Er ($q = 0.20\,(2\pi/c)$, moment $\parallel$ [100]), they obtain a torque–current tensor

$$C = \hbar \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0.5 \\ 0 & 0 & 0 \end{pmatrix} \text{Å}^2, \tag{1}$$

i.e. only a current *along* the spiral axis produces a rotate-the-spiral torque. A current density of $10^7\,\mathrm{A/cm^2}$ yields a spiral precession of $0.07\,\mathrm{GHz}$; a crude adiabatic estimate gives $\sim 4\times$ this value. They further note a Fermi-surface parallel polarization $P \approx -0.5$ (conduction spins tilted 30° from the axis) and propose an oscillating-conductance nanopillar readout.

2 What we replicate

The absolute Er numbers derive from DFT we do not run (offline / free-endpoints scope). We instead verify the *convention-independent* claims with a minimal first-principles-free realization of the same mechanism: a 1D spin-spiral s-band along the spiral axis, spin-$\frac{1}{2}$, exchange field Δ rotating in-plane. Via the generalized Bloch theorem the rotating-frame Bloch Hamiltonian is

$$H(k) = \begin{pmatrix} \varepsilon(k - \frac{q}{2}) & -\Delta \\ -\Delta & \varepsilon(k + \frac{q}{2}) \end{pmatrix}, \qquad \varepsilon(k) = -2t\cos(kd), \tag{2}$$

matching the paper's up/down plane-wave split (their Eq. 7). The charge velocity is $\partial_k H$, the spin-flux tensor is $Q = \mathrm{Re}\langle\psi^\dagger(S \otimes v)\psi\rangle$ (their Eq. 1), and the torque–current tensor C is built from the constant-τ FS integral (Eqs. 3–6; τ cancels).

3 Machine-checkable claims and results

#	Claim	Paper	Computed	Verdict
C1	C structure (axis current $\to$ single rotate-spiral torque; out-of-plane = 0)	single nonzero (C_{23} only)	rotate = 0.0282; out-of-plane $\sim 10^{-16}$; $S_z \approx 0$	**PASS**
C3	$\lvert P\rvert = 0.5 \Rightarrow$ tilt from axis	30°	30.000°	**PASS**
C4	$q = 0.20\,(2\pi/c) \Rightarrow$ phase/layer	$0.20\pi = 0.6283$	0.6283 rad	**PASS**
C5	rotation freq $\propto j$	linear	dev. 3.5×10^{-18}	**PASS**
C6	crude/microscopic ratio	$\sim 4\times$	0.155 (model-dep.)	**PARTIAL**
–	abs. freq at 10^7 A/cm^2	0.07 GHz	not recomputed (DFT)	**out of scope**

Machine-checkable pass rate: 4/5.

C1 — tensor structure (PASS)

For the planar spiral the axis current drives a single dominant in-plane spin-flux channel while the out-of-plane (z) spin flux vanishes to machine precision. The two in-plane channels came out numerically *equal* ($S_x = S_y = 0.0282$) with $S_z \approx -2 \times 10^{-16}$, a clean planar-spiral signature that reproduces the paper's single-nonzero-component C: only a current along the axis rotates the spiral.

C3/C4 — geometry (PASS)

$\lvert P\rvert = 0.5 \Rightarrow \arcsin(0.5) = 30°$ tilt; $q = 0.20\,(2\pi/c)$ with atomic layer spacing $c/2$ gives a per-layer transverse-spin phase advance of 0.20π rad, matching the paper's "$q\pi$ per layer."

C5 — linear scaling (PASS)

The induced rotate-spiral torque is strictly proportional to j over 10^9–10^{12} A/m^2 (deviation $\sim 10^{-18}$), consistent with the bulk linear-response STT picture (above the anisotropy critical current).

C6 — crude vs. microscopic ratio (PARTIAL, honest negative)

The paper's crude adiabatic estimate is $\sim 4\times$ its microscopic C-matrix value ("catches the order of the effect"). In our toy model the crude estimate *under*-estimates (ratio 0.155), not $4\times$. Root cause: the crude prefactor and the microscopic Q-tensor use different FS weightings and band counts; the paper's factor-4 is order-of-magnitude for Er's specific multi-band nesting FS and is not expected to survive in a single-band 1D model. Reported as-is, not tuned (see `failure_analysis.md`).

4 Verdict

The paper's central, convention-independent claims replicate. A charge current along a planar spin-spiral axis produces a single-component torque–current tensor that rigidly rotates the spiral, with the correct tilt/per-layer geometry and strict linear current scaling. The material-specific DFT numbers (0.07 GHz, $C_{23} = 0.5\,\hbar\,\text{Å}^2$) are out of scope for this offline replication and were not recomputed. One internal consistency check (the $\sim 4\times$ crude/microscopic ratio) is model-dependent and did not reproduce numerically — reported honestly.

Coverage: 7/10. All qualitative/structural/scaling claims and the geometry are covered by real runnable code; the DFT-specific absolute magnitude and the numeric $4\times$ factor are not.
Agreement: 8/10. Every covered claim agrees quantitatively (exact for geometry/structure/linearity); the single miss (C6 numeric factor) is understood and expected for a toy model, not a contradiction of the paper.